

$$u(x)$$

$$u''(x) = 0$$

$$u(x) = C_1 x + C_2 \cdot 1$$

$$u(x, y)$$

$$u_{xy} = 0$$

$$u = u_x : v_y = 0 \quad v = C(x)$$

$$u_x = C(x)$$

$$u = \int C(x) dx + C_1(y)$$

$\downarrow f(x) \quad \downarrow g(y)$

$$u(x, y) = f(x) + g(y)$$

Исследовать ДУ 2^{го} порядка с coeff. пр-м

$$Q \subset \mathbb{R}^n$$

$$x = (x_1, \dots, x_n) \in Q$$

$$u(x, y)$$

$$a_{ij}(x) = a_{ji}(x)$$

$$\forall i, \forall j = 1, \dots, n$$

$$L(u) := \sum_{i,j=1}^n a_{ij}(x) u_{x_i x_j} + \sum_{j=1}^n a_j(x) u_{x_j} + a(x) u = F(x) \quad (1)$$

$\swarrow \quad \searrow \quad \nearrow$
 $C(Q) \quad C(Q) \quad C(Q)$

$u(x)$ есть рещ. (1):

$$1) u(x) \in C^2(Q)$$

$$2) \text{ бнн (1) } \forall x \in Q$$

$$\begin{pmatrix} a_{11}(x) & \dots & a_{1n}(x) \\ \vdots & & \vdots \\ a_{n1}(x) & \dots & a_{nn}(x) \end{pmatrix} = A(x), \quad A^T(x) = A(x)$$

$$u_{x_i x_j} = u_{x_j x_i}$$

$$\begin{aligned} a_{ij} u_{x_i x_j} + a_{ji} u_{x_j x_i} &= (a_{ij} + a_{ji}) u_{x_i x_j} = \\ &= 2 \frac{a_{ij} + a_{ji}}{2} u_{x_i x_j} = \frac{a_{ij} + a_{ji}}{2} u_{x_i x_j} + \frac{a_{ij} + a_{ji}}{2} u_{x_j x_i} \end{aligned}$$

Итак упр-я б можно

$$x^0 \in Q \mapsto A(x^0) = \begin{pmatrix} a_{11}(x^0) & \dots & a_{1n}(x^0) \\ \vdots & & \vdots \\ a_{n1}(x^0) & \dots & a_{nn}(x^0) \end{pmatrix}$$

$$\xi = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}$$

$$(A(x^0) \xi, \xi) = \sum_{i,j=1}^n a_{ij}(x^0) \xi_i \xi_j = A_1(x^0) (\xi_1)^2 + \dots + A_n(x^0) (\xi_n)^2$$

$$\{A_1(x^0), \dots, A_n(x^0)\}$$

$$n_0 = \text{число нулевых} = n_0(x^0)$$

$$n_+ = \text{число положительных} = n_+(x^0)$$

$$n_- = \text{число отрицательных} = n_-(x^0)$$

$$n_0 + n_+ + n_- = n$$

$$\text{I } n_0 = 0$$

$$a) \begin{cases} n_+ = n \\ n_- = 0 \end{cases} \text{ или } \begin{cases} n_- = n \\ n_+ = 0 \end{cases}$$

$$\delta) \begin{cases} n_+ = n-1 \\ n_- = 1 \end{cases} \text{ или } \begin{cases} n_+ = 1 \\ n_- = n-1 \end{cases}$$

$$b) \begin{cases} n_+ \geq 2 \\ n_- \geq 2 \end{cases}$$

Ia эллиптическое в x^0

$u(x, y)$

$$\Delta u \begin{cases} u_{xx} + u_{yy} = 0 \\ 1 \cdot \xi_1^2 + 1 \cdot \xi_2^2 \end{cases}$$

ур. Лапласа

$$u(x_1, \dots, x_n) \quad u_{x_1 x_1} + \dots + u_{x_n x_n} = 0$$

$$\Delta u = 0$$

Ib гиперболическое в x^0

волновое ур-е

$u(x, t)$

$$u_{tt} = a^2 u_{xx}$$

$$a^2 u_{xx} - u_{tt} = 0$$

$$a^2 \xi_1^2 - 1 \cdot \xi_2^2$$

$u(x_1, x_2, x_3, t)$

$$u_{tt} = a^2 (u_{x_1 x_1} + u_{x_2 x_2} + u_{x_3 x_3})$$

$$u_{tt} = a^2 \Delta u$$

Ib ультрагиперболическое в x^0

$$u_{x_1 x_1} + u_{x_2 x_2} - u_{x_3 x_3} - u_{x_4 x_4} = 0$$

II $n_0 > 0$

параболическое в x^0

ур-е теплопроводности

$u(x, t)$

$$u_t = a^2 u_{xx}$$

$$a^2 u_{xx} - u_t = 0$$

$$a^2 \xi_1^2 + 0 \cdot \xi_2^2$$

$$u(x_1, x_2, x_3, t) \quad u_t = a^2 \Delta u$$

Понятие x -кн (харак-н пов-ми) ур-я (1)

S - $(n-1)$ -мерная пов-ть, $S \subset Q$

$$S: S(x) = 0$$

$$S(x) \in C^1$$

$$\bar{\partial} S(x) \neq \bar{0}$$

$$x^0 \rightarrow x^0 \in S \text{ (т.е. } S(x^0) = 0)$$

$$x^0 \rightarrow \sum_{i,j=1}^n a_{ij}(x^0) \frac{\partial S}{\partial x_i}(x^0) \cdot \frac{\partial S}{\partial x_j}(x^0) = 0$$

x^0 наз. характеристической для ур-я (1)

Если ВСЯ пов-ть S сост. из хар-ных точек, то пов-ть S наз. хар. пов-ть (или характеристикой) для ур-я (1)

$$S: S(x) = 0$$

$$S(x) = \text{const}$$

Суртуй $\mathbb{R}^2$

$$u(x, y)$$

$$(1) \rightarrow a(x, y)u_{xx} + 2b(x, y)u_{xy} + c(x, y)u_{yy} + k(x, y)u_x + l(x, y)u_y + h(x, y)u = F(x, y)$$

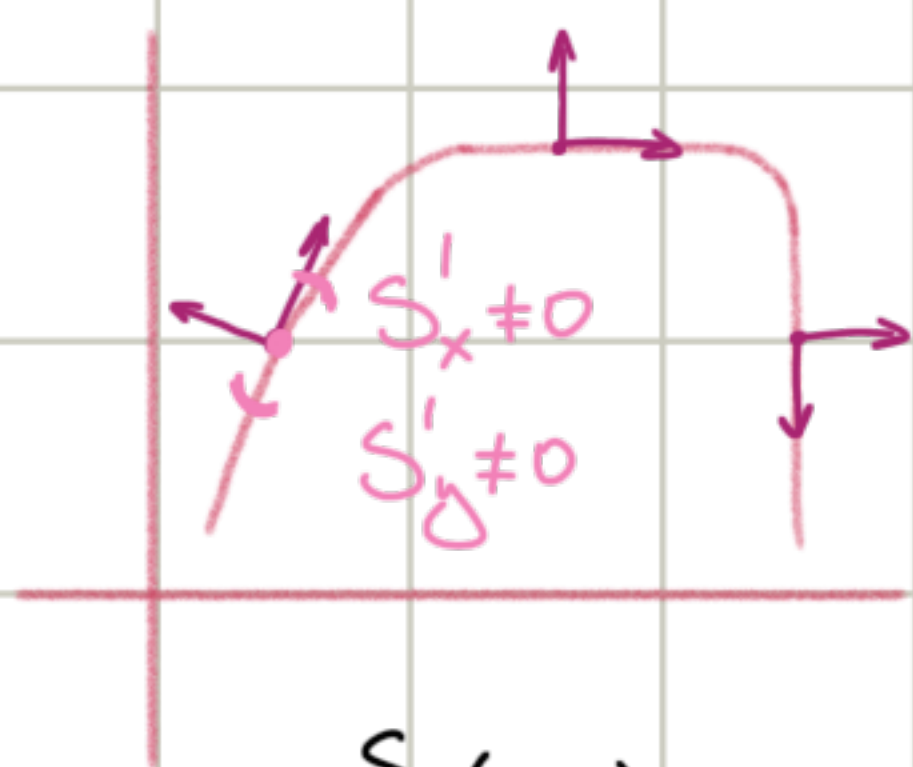
S-кривая на ну-ну

$$S: S(x, y) = \text{const}$$

$$S(x, y) \in C^1, \bar{\partial} S \neq 0$$

1) умо $S: S(x, y) = \text{const}$ емб x-ка y -а (1)

$$a(x, y) \left(\frac{\partial S}{\partial x} \right)^2 + 2b(x, y) \frac{\partial S}{\partial x} \frac{\partial S}{\partial y} + c(x, y) \left(\frac{\partial S}{\partial y} \right)^2 = 0 \quad | : \left(\frac{\partial S}{\partial y} \right)^2$$



$$S(x, y) = 0 \quad y = y(x)$$

$$S(x, y(x)) = 0$$

$$S'_x \cdot 1 + S'_y \cdot y'(x) = 0$$

$$y'(x) = -\frac{S'_x}{S'_y}$$

$$a(x, y) \frac{(S'_x)^2}{(S'_y)^2} + 2b(x, y) \frac{S'_x}{S'_y} + c(x, y) = 0$$

$$a(x, y) (y'(x))^2 - 2b(x, y) y'(x) + c(x, y) = 0$$

$$u_{xx} + u_{yy} = 0$$

$$y = y(x): (y'(x))^2 + 1 = 0 \quad \emptyset$$

$$u(x, t)$$

$$u_{tt} = a^2 u_{xx}$$

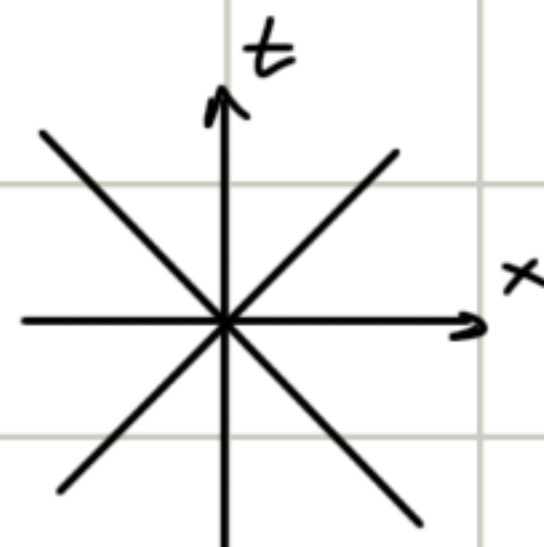
$$a^2 u_{xx} - u_{tt} = 0$$

$$t = t(x): a^2 (t'(x))^2 - 1 = 0$$

$$\begin{cases} t'(x) = \frac{1}{a} \\ t'(x) = -\frac{1}{a} \end{cases}$$

$$t = \frac{1}{a} x + C_1$$

$$t = -\frac{1}{a} x + C_2$$



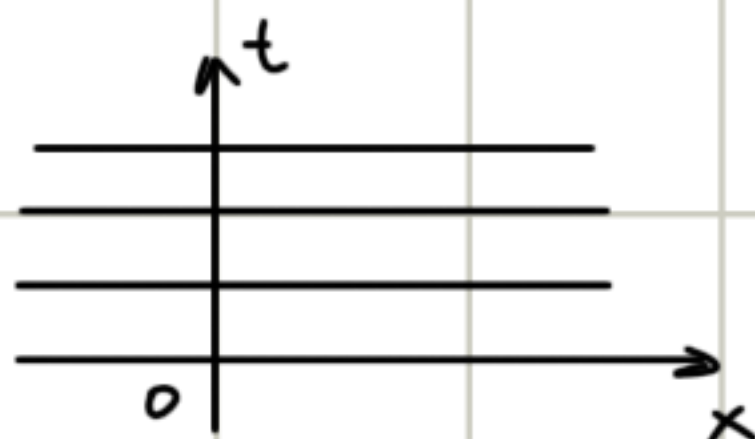
$$u_t = a^2 u_{xx}$$

$$a^2 u_{xx} - u_t = 0$$

$$a^2 (t'(x))^2 = 0$$

$$t'(x) = 0$$

$$t = \text{const}$$



$$y' = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\mathbb{R}^2 !!$ και βο χαρ-κ

0 $\longleftrightarrow$ ελλειψ.
2 $\longleftrightarrow$ υπερβ.
1 $\longleftrightarrow$ παραβ.

$$x + at = \tilde{z}_2$$

$$x - at = \tilde{z}_1$$

$$\begin{cases} \xi = x + at \\ \eta = x - at \end{cases}$$

$$u_x = u_{\xi} \xi_x + u_{\eta} \eta_x = u_{\xi} \cdot 1 + u_{\eta} \cdot 1$$

$$u_t = u_{\xi} \xi_t + u_{\eta} \eta_t = u_{\xi} a + u_{\eta} (-a)$$

$$u_{xx} = u_{\xi\xi} + u_{\eta\eta} + 2u_{\xi\eta}$$

$$\begin{matrix} \cdot a^2 \\ \cdot (-1) \end{matrix} + u_{\eta\eta} = 0$$

$$u_{tt} = u_{\xi\xi} a^2 + u_{\eta\eta} a^2 - 2a^2 u_{\xi\eta}$$

$$u_{\xi\xi} (a^2 - a^2) + u_{\eta\eta} (a^2 - a^2) + u_{\xi\eta} (2a^2 + 2a^2) = 0$$

$$u_{\xi\eta} = 0 \quad f(\xi) + g(\eta)$$

$$u(x, t) = f(x + at) + g(x - at)$$

$$u_{tt} = a^2 u_{xx}$$

$$\forall f(\xi) \in C^2, \forall g(\eta) \in C^2$$